BUSINESS REQUIREMENTS DOCUMENT

IT Help Desk Analytics & SLA Breach Prediction

Team Project | Version 1.0

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Document Control

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Revision History

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1.0	17 October 2026	Initial BRD prepared for project planning.	Project Team

1. Executive Summary

IT Help Desk Analytics & SLA Breach Prediction

The solution is intended to support prioritisation and operational decision-making. It will provide visibility into ticket volume, resolution time, backlog, priority, category, assignment group, SLA status, and historical breach patterns. Predictions will be decision-support signals only; they will not automatically change ticket priority, assign staff, or close tickets.

2. Business Background and Problem Statement

IT help desks receive incidents and service requests from employees or customers. Tickets differ by severity, category, assignment group, and required response or resolution time. When ticket information is spread across records or reviewed manually, it can be difficult to identify bottlenecks, understand repeat issues, and intervene before an SLA deadline is missed.

The project addresses the need for a consolidated view of help desk performance and a data-driven way to flag tickets that may breach their applicable SLA. The analysis will use available historical ticket data and clearly documented business rules.

3. Business Objectives

- Measure ticket volume, status, backlog, response time, and resolution time.
- Measure SLA compliance and breach rates using a consistent, documented definition.
- Identify patterns by priority, issue category, assignment group, channel, and time period where those fields are available.
- Highlight recurring causes of delay and operational bottlenecks.
- Develop and evaluate a predictive model to estimate SLA-breach risk for eligible tickets.
- Present findings through clear, interactive dashboards and concise recommendations.
- Provide a reproducible workflow for data cleaning, analysis, modelling, and reporting.

4. Project Scope

4.1 In Scope

- Importing an approved historical IT help desk ticket dataset.
- Data profiling, validation, cleaning, and documentation of missing, duplicate, inconsistent, and invalid records.
- Exploratory data analysis (EDA) and KPI calculation.
- Analysis of ticket trends, workload, resolution performance, SLA outcomes, and breach patterns.
- Dashboard development with appropriate filters and visualisations.
- Predictive modelling for SLA-breach risk, subject to sufficient suitable historical data and a valid target label.
- Model evaluation using classification metrics and a documented train/test strategy.
- Recommendations for monitoring, prioritisation, and process improvement.
- Final project documentation and presentation.

4.2 Out of Scope

- Replacing or modifying a production IT Service Management (ITSM) platform.
- Connecting to live company systems unless access is separately approved.
- Automatically assigning, escalating, closing, or reprioritising real tickets.
- Sending real-time alerts or notifications unless explicitly added and approved.
- Guaranteeing that every predicted ticket will breach or will not breach an SLA.
- Making staffing, disciplinary, or employee-performance decisions from model outputs alone.
- Predicting SLA breach where required labels, timestamps, or SLA rules are unavailable or unreliable.

5. Stakeholders and Users

Stakeholder / User	Interest or Responsibility
Business sponsor / project guide	Reviews objectives, scope, requirements, and acceptance.
IT Help Desk Manager	Uses KPI trends and breach-risk insights to support operational decisions.
Service Desk Analyst	Reviews ticket details, workload, and risk indicators to prioritise follow-up.
IT Support Team / Assignment Group	Uses insights to identify bottlenecks and recurring issue categories.
Data Analyst	Profiles data, defines KPIs, analyses trends, and develops dashboards.
Data Scientist / ML Team Member	Builds and evaluates the SLA-breach prediction model.
Project Team	Coordinates data preparation, development, validation, documentation, and presentation.
Data Owner / ITSM Administrator	Confirms field meanings, access permissions, SLA rules, and data quality.

6. Business Process Overview

– Ticket creation: a user reports an incident or requests a service; the ticket is recorded with available attributes.

– Ticket assignment: the ticket is assigned a priority, category, owner, or assignment group according to the source process.

– Service handling: support staff work on the ticket and update its status and timestamps.

– SLA monitoring: elapsed time is compared with the applicable response or resolution target, taking documented pause rules into account if available.

– Analytics: historical tickets are cleaned and analysed to calculate operational KPIs and identify patterns.

– Risk prediction: for eligible open tickets, the model estimates the likelihood of an SLA breach using information available at the prediction time.

– Human review: analysts use the dashboard and risk flag to decide whether follow-up or escalation is appropriate.

7. Business Requirements

ID	Requirement	Priority	Acceptance Evidence
BR-01	The solution shall consolidate the approved ticket data into an analysis-ready dataset.	Must	Source-to-output row counts and data preparation log are available.
BR-02	The solution shall report ticket volume and status distribution for the selected period.	Must	Dashboard totals reconcile with validated source records.
BR-03	The solution shall report SLA compliance and breach outcomes using approved SLA definitions.	Must	Definitions and calculation logic are documented and validated.
BR-04	The solution shall analyse resolution time and, where available, first-response time.	Must	Measures are calculated from valid timestamps and clearly labelled.
BR-05	The solution shall allow analysis by relevant dimensions such as priority, category, group, and date.	Must	Available dimensions can be filtered or compared in the report.
BR-06	The solution shall identify common ticket drivers and possible delay patterns.	Should	Findings are supported by charts or summary tables.
BR-07	The solution shall estimate SLA-breach risk for eligible tickets using a validated predictive model.	Must	Model evaluation results and limitations are documented.
BR-08	The solution shall present risk information in a way that supports human review.	Must	Risk score or risk band is labelled as an estimate, not a certainty.
BR-09	The solution shall document data quality issues, assumptions, and exclusions.	Must	Data dictionary and cleaning notes are included.
BR-10	The solution shall provide actionable findings and recommendations.	Should	Final report links each recommendation to an observed finding.
BR-11	The solution shall protect sensitive ticket information and restrict access to approved users.	Must	Only authorised, appropriately minimised data is used in deliverables.
BR-12	The solution shall support repeatable refresh and model evaluation where the project environment permits.	Could	Refresh steps and versioned evaluation results are documented.

8. Functional Requirements

ID	Functional Requirement	Acceptance Criteria
FR-01	Data ingestion	The workflow imports the approved dataset and records the source and extraction date if known.
FR-02	Data validation	Required fields, data types, duplicate identifiers, timestamp order, and invalid values are checked.
FR-03	Data cleaning	Cleaning rules are applied consistently; records removed or transformed are logged.
FR-04	KPI calculation	The system calculates the agreed ticket, time, backlog, and SLA measures.
FR-05	Dashboard filtering	Users can filter supported visuals by available dimensions and date range.
FR-06	Trend analysis	The report displays ticket trends and relevant operational comparisons over time.
FR-07	SLA risk scoring	The model produces a breach-risk probability or risk band for eligible tickets, if the data supports it.
FR-08	Model evaluation	The workflow reports suitable classification metrics and compares model performance against a baseline.
FR-09	Interpretability	The project explains the main factors associated with risk, without implying causation unless established.
FR-10	Export / presentation	The project can present results in agreed dashboard, report, or presentation formats.
FR-11	Documentation	The project includes field definitions, KPI formulas, assumptions, model details, and known limitations.

9. Data Requirements

The exact field list must be confirmed against the dataset selected by the team. The fields below are recommended candidates, not a claim that every source contains them.

Field Group	Candidate Fields	Purpose / Validation
Ticket identity	Ticket ID / case ID	Unique record identification; investigate duplicates and missing identifiers.
Time fields	Created date/time, first response date/time, resolved date/time, closed date/time, SLA due date/time	Validate formats, time zones, chronological order, and whether timestamps are recorded consistently.
Ticket status	Status, reopened flag, cancelled flag	Define open backlog and determine which records are eligible for analysis.
Priority and type	Priority, incident/request type, category, subcategory	Standardise labels and assess performance by ticket characteristics.
Ownership	Assigned team/group, assignee, support tier	Analyse workload and hand-offs; remove or mask personal identifiers when unnecessary.
SLA fields	SLA type, target duration, SLA outcome, breach indicator, pause/exclusion duration	Confirm the official rule for response versus resolution SLA and any pause calendar.
Additional context	Channel, location/site, service, business impact	Use only when present, relevant, permitted, and sufficiently complete.
Derived fields	Resolution duration, response duration, age at scoring, time remaining, breach label	Derive using documented rules; prevent leakage from information not available at prediction time.

9.1 Data Quality Rules

- Check uniqueness of ticket identifiers and define how duplicate records are resolved.
- Measure missingness for each field; do not silently replace unknown values with arbitrary values.
- Validate timestamps and investigate resolution times that are negative or implausibly long.
- Standardise category, priority, status, and assignment-group labels without erasing meaningful distinctions.

- Separate cancelled, test, duplicate, and out-of-scope tickets according to approved rules.
- Record all exclusions and transformations so the analysis can be reproduced.
- Avoid using personal or confidential fields unless required and authorised.

10. KPI Definitions

The project team and data owner must approve these definitions before publishing final figures. KPI values depend on the source data and the agreed SLA policy.

KPI	Proposed Definition	Important Note
Total Tickets	Count of distinct in-scope ticket IDs.	Use the agreed reporting period and exclusions.
Open Backlog	Count of in-scope tickets not in a final/closed state at the reporting cut-off.	Define statuses considered open, pending, or resolved.
SLA Breach Count	Count of eligible tickets whose approved SLA outcome is breached.	Keep response SLA and resolution SLA separate if both exist.
SLA Breach Rate	Breached eligible tickets ÷ all eligible tickets with a determinable SLA outcome × 100.	Do not include tickets with unknown SLA outcome in the denominator.
SLA Compliance Rate	Compliant eligible tickets ÷ all eligible tickets with a determinable SLA outcome × 100.	Should complement breach rate when outcomes are strictly binary.
Average Resolution Time	Mean of valid resolved-time minus created-time durations.	Also consider median and percentiles because durations may be skewed.
Median Resolution Time	50th percentile of valid resolution durations.	State unit and treatment of reopened tickets.
First Response Time	First valid response timestamp minus created timestamp.	Calculate only when both timestamps are available and the definition is agreed.
Breach Risk	Model-estimated probability or risk category for SLA breach at a specified scoring time.	Not a confirmed outcome; threshold must be selected on validation data.
Model Recall for Breaches	True breached tickets predicted as breach ÷ all actual breached tickets in the evaluation set.	Important where missing a likely breach is costly.
Model Precision for Breaches	True breached tickets predicted as breach ÷ all tickets predicted as breach.	Indicates how often risk flags are correct.

11. SLA Breach Prediction Requirements

11.1 Prediction Objective

Estimate whether an eligible ticket will breach its applicable SLA, using only information available at the defined prediction time. The team must specify whether the target is response-SLA breach, resolution-SLA breach, or a single approved breach definition. These targets must not be combined without a clear business reason.

11.2 Target Label and Eligibility

- Define a positive label (1) as an SLA breach and a negative label (0) as an SLA met outcome, based on the approved source rule.
- Exclude tickets with unknown or indeterminate target outcomes from supervised training and final evaluation unless a justified method is approved.
- Define the population being scored, such as open tickets that have not yet reached the SLA outcome.
- Confirm the SLA target, business calendar, pauses, reassignment rules, and time-zone handling with the data owner.

11.3 Candidate Predictors

Potential predictors may include priority, category, channel, assignment group, creation hour/day, ticket age at scoring, elapsed time, time remaining to the SLA deadline, prior workload measures, and historical group-level performance. Only features available at prediction time and permitted for use may be included.

11.4 Data Leakage Controls

- Do not use final resolution/closure timestamps, final SLA outcome fields, or other post-outcome information as predictors.

- Do not calculate features using information that would not have been known at the ticket's scoring time.
- Split data by time where practical so evaluation better represents predicting future tickets; avoid placing near-duplicate records across splits.
- Fit imputers, encoders, scalers, feature selection, and resampling steps only on training data within the validation workflow.
- If multiple tickets belong to the same incident or linked case, consider grouped splitting to prevent information leakage.

11.5 Model Development and Evaluation

- Start with a simple baseline, such as predicting the majority class or a transparent baseline classifier.
- Compare one or more suitable classification algorithms; select based on validation results and project constraints.
- Use precision, recall, F1-score, confusion matrix, and PR-AUC where appropriate; include ROC-AUC when informative.
- Review class imbalance and do not rely on accuracy alone if breaches are uncommon.
- Select the operational risk threshold using validation data and the relative cost of missed breaches versus unnecessary alerts.
- Report results on a held-out test set that was not used to choose features, tune parameters, or select the threshold.
- Document model version, selected features, split strategy, random seed where relevant, metrics, limitations, and intended use.

11.6 Model Output

Output	Description
Ticket identifier	Reference to the ticket record, using a non-sensitive identifier where possible.
Scoring timestamp	Time at which the risk estimate was generated.
Estimated breach probability	Probability estimate if the selected model supports a meaningful probability output.
Risk band	Optional Low / Medium / High band based on documented validation-based thresholds.
Reason indicators	Optional model explanation or contributing factors, described as associations rather than causes.
Caveat	The prediction is an estimate to support human review, not a guarantee or an automated decision.

12. Dashboard and Reporting Requirements

Dashboard Page / Section	Proposed Content
Overview	Total tickets, open backlog, SLA breach rate, compliance rate, average/median resolution time, and trends.
SLA Performance	Breach versus compliance, breach trends, and comparisons by priority, category, and assignment group.
Ticket Operations	Ticket volume by status, category, priority, channel, and assignment group; workload and aging where possible.
Resolution Analysis	Resolution-time distribution, first-response performance where available, and long-running ticket patterns.
Prediction & Risk	Eligible open tickets, risk distribution, high-risk list or summary, and model evaluation summary.
Data Quality & Definitions	Coverage, missingness, exclusions, KPI definitions, and important assumptions.

Visuals should use consistent labels, appropriate units, readable contrast, and filters that do not imply unsupported precision. Every chart should identify its reporting period and relevant denominator where needed.

13. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Accuracy	KPI calculations and SLA labels must follow approved definitions and reconcile to validated records.
NFR-02	Usability	Dashboard pages must be understandable to a non-technical help desk stakeholder.
NFR-03	Performance	The report should load and filter within reasonable time for the chosen dataset and environment.
NFR-04	Security & Privacy	Use authorised data, minimise personal data, and mask sensitive fields in screenshots and deliverables.
NFR-05	Reliability	The workflow should produce consistent results from the same source data and configuration.
NFR-06	Maintainability	Code, transformations, KPI logic, and modelling steps should be organised and documented.
NFR-07	Reproducibility	Document software versions, key parameters, data splits, and refresh instructions.
NFR-08	Explainability	Explain model purpose, major limitations, and risk interpretation in plain language.
NFR-09	Accessibility	Use readable fonts, adequate contrast, meaningful titles, and avoid colour as the only status cue.
NFR-10	Ethical use	Predictions must support human review and must not be used as the sole basis for punitive decisions.

14. Assumptions and Dependencies

14.1 Assumptions

- The team will obtain a permitted historical help desk ticket dataset, either anonymised real data or a clearly labelled synthetic/public dataset.
- The selected dataset contains sufficient records and reliable fields for the planned descriptive analysis.
- A supervised prediction model will be attempted only if an adequate, trustworthy SLA outcome label exists.
- Business definitions and SLA rules can be documented; if they cannot be confirmed, assumptions will be stated and the result labelled accordingly.
- The project is an academic/portfolio proof of concept and is not a production service.

14.2 Dependencies

- Dataset availability, permitted usage, and data dictionary.
- Confirmation of SLA rules and relevant business calendars.
- Availability of required tools, such as Python and a dashboarding tool.
- Agreement on project roles, delivery dates, and review checkpoints.
- Adequate examples of both breach and non-breach outcomes for model evaluation.

15. Risks and Mitigation

Risk	Potential Impact	Mitigation
Missing or unreliable SLA labels	Invalid breach-rate calculations and weak model target.	Validate definitions with data owner; disclose uncertainty; do not train on unknown labels.
Class imbalance	Model may overlook breaches while showing high accuracy.	Use recall, precision, F1, PR-AUC, confusion matrix, and baseline comparisons.
Data leakage	Overly optimistic test results that fail in practice.	Use time-aware splits and exclude post-outcome fields.
Insufficient dataset size	Unstable or non-generalizable results.	Use a simpler model, cross-validation where appropriate, and clearly state limitations.
Changing SLA policies	Historical labels may not match current rules.	Document policy periods and assess whether records should be segmented.
Privacy exposure	Sensitive ticket or employee information may be disclosed.	Minimise and mask data; use approved access and sanitized screenshots.
Misinterpretation of risk scores	Users may treat estimates as certainty.	Label scores clearly and retain human review.
Scope expansion	Delayed completion and inconsistent deliverables.	Prioritise must-have requirements and approve changes with the project guide.

16. Deliverables

- Approved Business Requirements Document (this document).
- Dataset description and data dictionary.
- Data quality and cleaning report.
- Reproducible data preparation and analysis code or workbook.
- Exploratory analysis and KPI summary.
- Interactive dashboard or agreed reporting output.
- SLA-breach prediction notebook/script and evaluation report, if data suitability permits.
- Final findings, recommendations, limitations, and project presentation.
- README or technical handover notes describing setup and execution.

17. Acceptance Criteria

- The dataset source, scope, fields, and data-quality rules are documented.
- Cleaning steps and exclusions are traceable and repeatable.
- KPI formulas and SLA definitions are reviewed and used consistently.
- Dashboard figures reconcile with the validated analysis dataset within documented rounding rules.
- Filters and visuals operate as intended for the supported dimensions.
- If prediction is feasible, the model is evaluated on data not used for model selection and includes appropriate metrics and a leakage review.
- If prediction is not feasible, the reason is documented and the project still delivers descriptive SLA analytics.
- Conclusions and recommendations are supported by analysis, and limitations are stated.
- No sensitive or unauthorised information appears in the final report, dashboard screenshots, or presentation.
- The project guide / designated reviewer accepts the final deliverables.

18. Proposed Project Phases

Phase	Main Activities	Exit Criteria
1. Discovery	Confirm business questions, users, scope, dataset, and SLA definitions.	Scope and definitions agreed.
2. Data Understanding	Profile fields, assess data quality, inspect labels and timestamp coverage.	Data suitability and limitations documented.
3. Preparation	Clean data, engineer valid features, create analysis-ready dataset.	Validated dataset and transformation log available.
4. Analytics	Calculate KPIs, conduct EDA, identify trends and bottlenecks.	Findings checked against definitions.
5. Dashboard	Build visuals, filters, KPI cards, and explanatory notes.	Dashboard reviewed by team / guide.
6. Prediction	Create leakage-safe splits, train baseline and candidate models, evaluate and choose threshold.	Evaluation report completed or infeasibility documented.
7. Validation	Check calculations, usability, privacy, and model limitations.	Acceptance checklist completed.
8. Final Delivery	Prepare final report, presentation, and handover instructions.	Deliverables submitted and reviewed.

19. Open Questions Requiring Team Confirmation

The BRD avoids assuming a particular dataset or tool. The team should confirm the following before implementation:

- Which dataset will be used, and is it real, public, or synthetic?
- Does the dataset contain a reliable SLA outcome, due time, or response/resolution target?
- Is the prediction target response-SLA breach, resolution-SLA breach, or another formally defined outcome?
- At what point should risk be predicted: ticket creation, after assignment, or at a periodic checkpoint?
- Which dashboard tool and modelling environment will be used?
- What reporting period and ticket exclusions should be applied?
- Who will review and approve KPI definitions and final acceptance?

20. Glossary

Term	Meaning
IT Help Desk	A team or function that records and resolves IT incidents and service requests.
Ticket	A record representing an incident, issue, or service request.
SLA	Service Level Agreement: a defined service target, such as response or resolution time.
SLA Breach	Failure to meet the applicable SLA target under the agreed policy.
Backlog	Tickets that remain open or otherwise unfinished at a specified time.
EDA	Exploratory Data Analysis: examining data to understand distributions, quality, and patterns.
Target Label	The outcome a supervised model is trained to predict.
Data Leakage	Use of information in model development that would not legitimately be available at prediction time.
Precision	Among tickets predicted as breach, the proportion that actually breached.
Recall	Among tickets that actually breached, the proportion correctly flagged by the model.
Risk Score	An estimate of the likelihood of breach; not a guarantee of the outcome.

Approval

Role	Name	Signature / Confirmation	Date
Project Guide / Reviewer			
Team Representative			

End of Document